

General Assignment Guidelines

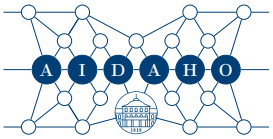
Please check the following list before handing in your mandatory assignments.

1. **Code:**
   Your code should be working and user-friendly. This means (for example):

   - Your code must be working. Submissions containing syntactic or runtime errors will not be debugged and may receive zero credit. If you change something in your code, run it again before handing it in.
   - Your code should be user-friendly. Use comments to explain to the user of your script what you are doing, what the respective objects are, which output is created etc.
   - Your code must be directly usable. The only adjustment we are going to make is changing a working directory or a path variable.
   - As a rule, the user of your program should not be required to change anything in order to run it successfully.
   - Your code must be reproducible. All required packages, data, and dependencies must be clearly specified. If necessary, include instructions for installing required packages.
   - If your code involves randomness (e.g. simulations), set a seed to ensure reproducibility of results.
   - Include a README file that briefly explains the structure of your repository and how to run your code.
   - The code must generate the results that are presented in the `.pdf`. This also refers to plots. If your `.pdf` includes a plot with a red time series, then running the code should not produce a graph with a blue time series. The result should be exactly the graph defined in your code.
   - You should split your code into multiple script files as indicated in the assignment. Keep function definitions separate. It can also be helpful if this helps your code to be neat. However, keep in mind that too many different script files will reduce the user-friendliness.

2. **Interpretation (`.pdf`):**

   - When writing your mandatory assignment, assume that its reader is an interested fellow student.
   - The file format of your mandatory assignment has to be `.pdf`. The use of LaTeX is recommended but not required (You find an example in the "Software" folder on ILIAS). Word files (`.docx`) are not acceptable.



- Include a title page that lists your name, study program, and student ID. If you are working with a partner, the title page has to include the information for both of you.
- Make sure you have a consistent layout throughout your assignment (e.g. do not change the font (size) in the middle of a sentence).
- When choosing the font size and especially when scaling graphs, make sure that they are easily readable when printing your document in A4 format. Neither LaTeX nor R will give you a warning when your graphs are too small to properly depict the information that is carried in them. You will find more detailed information on graphs on the next pages.
- Enumerate and name your graphs and tables. See that they are readable, nice looking, and have a high enough resolution.
- Ensure not to include more decimal places than necessary in your tables. Generally, **four decimal** places is considered adequate.
- Use captions to give information on the content of your graphs and tables. This means a graph or table consists of a meaningful title caption (not just Figure 1), a description below the title and the graph or table itself.
- Create formulas directly in LaTeX or use a formula editor to make them readable.
- If you refer to books or papers, cite them properly.
- Read your interpretation carefully before handing it in. Use a spell-checking software.
- Write your mandatory assignment as if it was a short paper. It should not be a description of tasks and of your R code.
- Keep the assignment short. No more than **five pages in total** (graphs, tables, formulas, and references included, title page excluded).
- Use references only when they are necessary.
- A suitable and professional font type for your paper is a serif font.

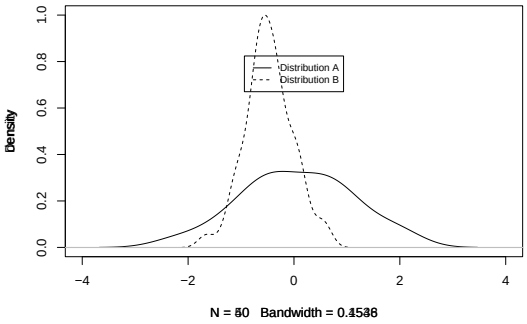
3. **Graphs:**
   The following lines show some of the possible mistakes that can occur when creating graphs. Figure 1 consists of three plots, each of which depicts two kernel densities. In Figure 1a, the position of the legend hides interesting parts of one density and the plot is printed very small. Also, the standard labels of the axes is print on top of each other as we print two kernel densities with two `plot()` calls in one image. In Figure 1b, the legend is printed partly outside the box and the plot is rather small as well. Also, the densities are not aligned properly, the zero is not the same for both graphs. Hence, both

graphs are not nice looking. Figure 1c uses a sensible position of the legend, is printed larger, and the densities are clearly visible. The labels are printed only once and the alignment is corrected. This is a nice graph.

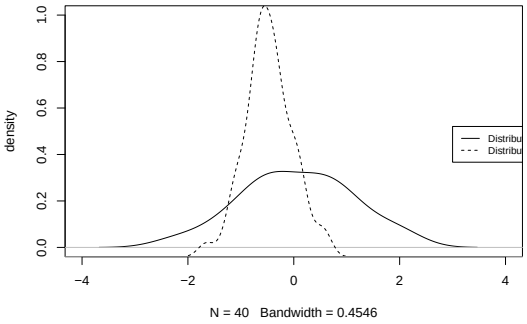
Abbildung 1: Positioning legends

Figures 1a-1c illustrate the same kernel densities with different color options and legends.

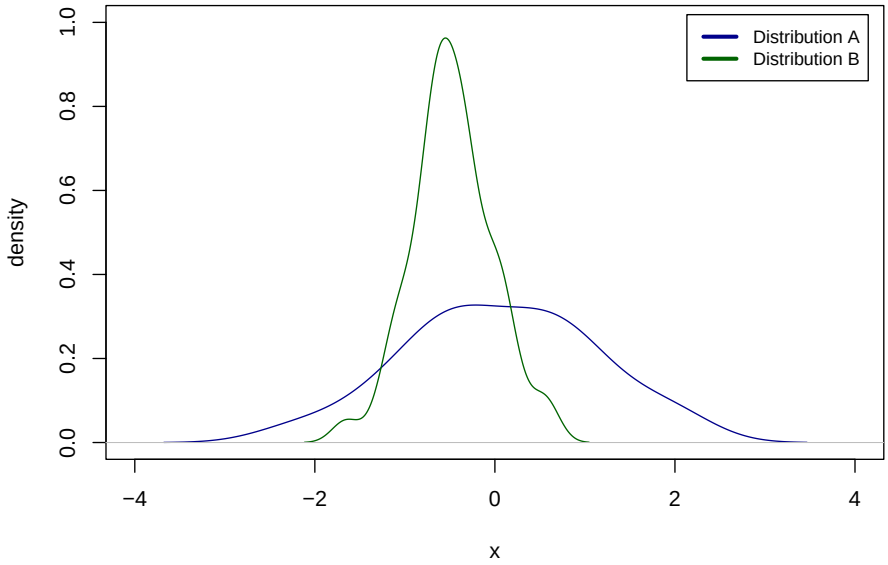
(a) not nice

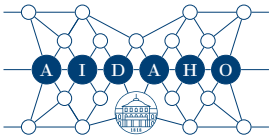


(b) not nice



(c) nice





4. **Upload**

   Your assignment should be handed in via a git repository. One repository per team! Use the structure of the git repository indicated in the first task of the assignment.

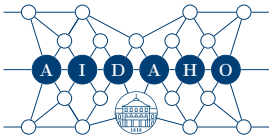
   - The used data should be saved in the folder `01_data`.
   - All your R scripts should be saved in `02_code/R`.
   - Your written report should be saved in `03_report`.
   - All graphs that you display in your report should be additionally saved in `03_report/graphs`.
   - Before the assignment deadline add Dr. Johannes Bleher (jbleher), and Jan Dvorak (jdvorak) as members (role: owner) to your git repository.
   - The assignment requires the implementation of a functioning WebApp. Please provide its URL in the written report and in the repository README. The WebApp must be accessible at the time of grading; if it is unavailable or non-functional, this may negatively affect the grade.
   - We will grade the latest commit on the default branch (usually `main`) that was pushed before the assignment deadline.
   - **Additionally, to validate your submission:** Upload **only** your written report (`.pdf` file) on ILIAS. If you work as a team of two, **both team members** have to upload the report on ILIAS.

5. **Use of Generative AI (e.g. ChatGPT, Copilot):** The use of generative AI tools is permitted and encouraged as part of your workflow. The effective and critical use of AI tools is considered part of modern data analysis practice. However:

   - You must clearly document where and how such tools were used (e.g. code generation, debugging, text editing). Include a brief statement (max. 5 lines) at the end of your report describing your use of AI tools.
   - You remain fully responsible for the correctness, understanding, and originality of your submission.
   - Blindly copying generated code or text without verification is not acceptable.
   - If requested, you must be able to explain and reproduce all parts of your solution without AI assistance.

6. **Grading**

   Please be aware that assistance on the take home assignments is only provided during the respective feedback sessions. A completed assignment consists of a `.pdf` file, the working R script and other code files including all datasets that are needed to run it, as well as a running web service on your assigned server – as laid out in the assignment.



All parts of your assignment should be stored in a git repository on the AIDAHO-Gitlab.

This assignment makes up 30% of your final grade and is mandatory. You will need to pass the assignment in order to pass the course. The language of the take home assignments is English.

You can update your solutions on the AIDAHO-Gitlab and on ILIAS as long as the deadline has not passed. Cheating (e.g., submitting somebody else's take home assignment, copying, etc.) will result in failing the take home assignment and, thus, the course.

Grading is based on:

- A correct and working code
- An accessible web service
- User-friendliness of the code (in particular: commenting)
- The quality of your interpretation
- The quality of your graphs
- The form (language, format etc.)

If you have any questions, send an e-mail to: `johannes.bleher@uni-hohenheim.de`